

Physiologically-Constrained Neural Operators

for Heart Sound Classification

A Complete Expert Guide from First Principles

Fourier

Neural Operator · PhysioFrequencyMask ·
PhysioFeatureExtractor · CNN Baseline

“We encode

known structural properties of cardiac acoustics as architectural inductive biases, analogous to physics-consistent neural architectures in continuum mechanics.”

Designed for readers with no prior background.

Every concept is derived from first principles. Every architectural choice is justified.

- Cardiac anatomy and the physics of heart sounds
- Fourier analysis from scratch: DFT, FFT, power spectrum
- Neural operators: from convolution to FNO
- PhysioFrequencyMask: per-mode logits + learnable band boundaries
- PhysioFeatureExtractor: 18 physical quantities fully derived
- Hilbert transform, analytic signal, instantaneous frequency
- Cross-beat consistency and harmonic analysis
- Structural vs. penalized physics constraints
- CNN baseline: STFT, Mel filterbank, BatchNorm
- Loss functions, optimizers, learning rate schedules
- t-SNE, confusion matrices, clinical interpretation
- Connection to PINNs, polyconvexity, continuum mechanics

Notebook: `heart_sound_fno_classifier.ipynb`

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1. Clinical Background: The Heart, Its Sounds, and Why They Matter

1.1 Cardiac Anatomy in Brief

The heart is a four-chambered muscular pump. The **left ventricle** (LV) ejects oxygenated blood into the aorta; the **right ventricle** (RV) pumps deoxygenated blood to the lungs. Blood flow through the heart is controlled by four valves:

Valve	Location	Function
Mitral (MV)	Left atrium → left ventricle	Opens in diastole, closes at start of systole
Tricuspid (TV)	Right atrium → right ventricle	Same, right side
Aortic (AV)	Left ventricle → aorta	Opens in systole, closes at end of systole
Pulmonary (PV)	Right ventricle → pulmonary artery	Same, right side

The cardiac cycle alternates between **systole** (ventricular contraction, ejection) and **diastole** (ventricular relaxation, filling). One cycle typically lasts 0.6–1.0 s at rest.

1.2 The Physical Origin of Heart Sounds

Heart sounds are acoustic pressure waves generated by rapid deceleration of blood and vibration of cardiac structures at valve closure. They are transmitted through the chest wall and recorded by a microphone as the **phonocardiogram (PCG)**.

Timing: At the beginning of systole.
Cause: Abrupt closure of the mitral and tricuspid valves. The sudden deceleration of blood creates a pressure transient that vibrates the valve leaflets, papillary muscles, and ventricular walls.
Frequency content: Predominantly 25–45 Hz. Low-frequency because the structures involved (thick muscular walls, large leaflets) are mechanically massive and vibrate slowly.
Duration: ≈ 100–170 ms.
Clinical significance: Louder in tachycardia and mitral stenosis; softer in first-degree heart block and mitral regurgitation.

Timing: At the end of systole / beginning of diastole.
Cause: Abrupt closure of the aortic and pulmonary valves. These are semilunar valves (smaller, stiffer than the atrioventricular valves), producing a higher-frequency sound.
Frequency content: Predominantly 50–70 Hz. Higher than S1 because the semilunar valves are smaller and stiffer.
Duration: ≈ 60–100 ms.
Clinical significance: Normally split into A2 (aortic) and P2 (pulmonary) compo-

nents.

Murmurs arise from **turbulent blood flow**. In laminar flow, fluid moves in smooth parallel layers and produces no sound. When flow velocity exceeds a critical threshold (Reynolds number $Re = \rho v D / \mu > \sim 2000$), turbulence develops, generating broadband acoustic energy.

Frequency content of murmurs: Extends from the S1/S2 range up to 400–600 Hz and beyond, depending on the severity and nature of the lesion.

- **Aortic Stenosis (AS):** Narrowed aortic valve creates high-velocity jet through the stenotic orifice during systole. Produces a *crescendo-decrescendo systolic ejection murmur* peaking at mid-systole. Frequency: 100–500 Hz.
- **Mitral Regurgitation (MR):** Incompetent mitral valve allows backward flow from LV to left atrium during systole. Produces a *holosystolic murmur* (entire systole). Frequency: 100–400 Hz.
- **Mitral Stenosis (MS):** Narrowed mitral valve restricts filling during diastole. Produces a *low-pitched diastolic rumble*. Frequency: 50–200 Hz, often overlapping the S1 band.
- **Mitral Valve Prolapse (MVP):** Billowing of mitral leaflets into the left atrium during systole. Characteristic *mid-systolic click* followed by a late systolic murmur.

1.3 One Cardiac Cycle in Time

A single cardiac cycle at 75 BPM ($T = 0.8$ s) unfolds as follows:

Phase	Duration	Acoustic event
Isovolumetric contraction	~ 50 ms	S1: MV + TV closure
Systolic ejection	~ 300 ms	Ejection murmur (if AS/MR)
Isovolumetric relaxation	~ 50 ms	S2: AV + PV closure
Diastolic filling	~ 400 ms	Diastolic murmur (if MS); S3/S4 if present

The S1-to-S2 interval is approximately 300 ms (systole duration) and the S2-to-next-S1 interval is approximately 500 ms (diastole duration) at rest. This asymmetry between systole and diastole is a key physiological feature the model must learn.

1.4 Why Automated PCG Classification?

- **Skill barrier:** Accurate cardiac auscultation requires 3–5 years of clinical training. Studies show inter-physician agreement on murmur classification of only 40–70%.
- **Access:** Over 2 billion people lack access to specialist cardiac care. An automated system embedded in a smartphone-connected stethoscope could enable mass screening.
- **Consistency:** Unlike a human, an algorithm applies the same criteria every time, enabling longitudinal tracking of a patient’s condition.

2. Signal Processing Foundations

2.1 What Is a Signal?

A **signal** is a function $s : \mathbb{R} \rightarrow \mathbb{R}$ (or $s : \mathbb{Z} \rightarrow \mathbb{R}$ for discrete signals) that carries information. In our case, $s(t)$ is the acoustic pressure at the chest wall microphone as a function of time t . A PCG signal is:

- **Real-valued:** pressure is a scalar at each moment.
- **Bandlimited:** meaningful cardiac content lies below ~ 1 kHz.
- **Non-stationary:** the statistical properties change over time (S1 is different from the silence after it).
- **Quasi-periodic:** the cardiac cycle repeats, but not perfectly.

2.2 Sampling and the Nyquist Theorem

A continuous signal $s(t)$ is sampled at discrete times $t_n = n/f_s$ to produce a discrete sequence $s[n] = s(n/f_s)$. The **sample rate** $f_s = 22,050$ Hz used here means 22,050 measurements per second.

Theorem 2.1 (Nyquist-Shannon Sampling Theorem). A bandlimited signal with maximum frequency $f_{\max}$ can be perfectly reconstructed from its samples if and only if $f_s \geq 2f_{\max}$ (the Nyquist criterion).

With $f_s = 22,050$ Hz, we can faithfully represent signals up to $f_{\max} = 11,025$ Hz. Since cardiac sounds are below 1 kHz, we have a comfortable margin.

2.3 The Fourier Transform: Decomposing Signals into Frequencies

Why the Fourier Transform?

The Fourier transform answers: *which frequencies are present in a signal, and how strongly?* For a periodic signal like a PCG recording, frequency-domain representations are often more informative than time-domain representations, because the clinically relevant features (S1 at 25–45 Hz, S2 at 50–70 Hz, murmurs at 100–500 Hz) are defined in terms of frequency.

2.3.1 Continuous Fourier Transform

The continuous Fourier transform of $s(t)$ is:

$$\hat{s}(\xi) = \int_{-\infty}^{\infty} s(t) e^{-2\pi i \xi t} dt$$

where ξ is frequency in Hz and $e^{-2\pi i \xi t} = \cos(2\pi \xi t) - i \sin(2\pi \xi t)$ is a complex exponential at frequency ξ . The inverse is:

$$s(t) = \int_{-\infty}^{\infty} \hat{s}(\xi) e^{2\pi i \xi t} d\xi$$

This tells us: $\hat{s}(\xi)$ is the “amplitude” (complex) of frequency ξ in s .

2.3.2 Discrete Fourier Transform (DFT)

For a finite sequence of L samples $s[0], s[1], \dots, s[L-1]$:

$$\hat{s}[k] = \sum_{n=0}^{L-1} s[n] e^{-2\pi i k n / L}, \quad k = 0, 1, \dots, L-1$$

The frequency corresponding to bin k is $f_k = k \cdot f_s / L$ in Hz. For $L = 66,150$ and $f_s = 22,050$ Hz, each bin represents $f_s / L \approx 0.333$ Hz.

2.3.3 Real FFT (rfft)

Since $s[n]$ is real, $\hat{s}[L-k] = \overline{\hat{s}[k]}$ (Hermitian symmetry). We only need to store bins $k = 0, 1, \dots, L/2$, giving $L/2 + 1$ unique complex values. PyTorch's `torch.fft.rfft` exploits this, and `rfftfreq` returns the corresponding frequencies $[0, f_s/L, 2f_s/L, \dots, f_s/2]$.

2.3.4 Power Spectral Density (PSD)

The PSD $P(\xi) = |\hat{s}(\xi)|^2 = (\text{Re } \hat{s})^2 + (\text{Im } \hat{s})^2$ measures the *energy* at each frequency. For a PCG signal:

- The PSD has large peaks in the S1 band (25–45 Hz) and S2 band (50–70 Hz).
- In AS, there is additional energy in the 100–500 Hz range during systole.
- In normal recordings, energy outside the cardiac bands is mostly noise.

2.3.5 The `norm='ortho'` Convention

In PyTorch, `rfft(x, norm='ortho')` applies a normalization factor of $1/\sqrt{L}$ to the forward transform and $1/\sqrt{L}$ to the inverse. This makes the transform **unitary**: $\|\hat{x}\| = \|x\|$ (Parseval's theorem). This is important in the FNO because it means spectral operations preserve energy scaling, preventing the values from growing or shrinking through FFT layers.

2.4 The Short-Time Fourier Transform (STFT)

A limitation of the standard DFT is that it treats the entire signal as stationary. But a PCG signal is *non-stationary*: S1 occurs at one time, silence at another, S2 at another. The STFT provides a time-frequency representation:

$$\mathcal{S}(m, k) = \sum_{n=0}^{N-1} s[n + m \cdot h] w[n] e^{-2\pi i k n / N}$$

where $w[n]$ is a **window function** of length N and h is the **hop length** (shift between successive windows). The Hann window $w[n] = \sin^2(\pi n / N)$ is used because it provides good frequency resolution while suppressing spectral leakage at window edges.

2.4.1 Spectral Leakage and Why Windowing Matters

If we abruptly cut a signal (rectangular window), the discontinuity at the edges creates artificial high-frequency components in the DFT — this is **spectral leakage**. The Hann window tapers to zero at both ends, suppressing this effect at the cost of slightly reduced frequency resolution. For cardiac signals, suppressing leakage is more important than maximal resolution.

2.5 The Mel Scale and Perceptual Frequency

The human auditory system does not perceive frequencies linearly. Our pitch perception is approximately logarithmic: the difference between 100 Hz and 200 Hz sounds as large as the difference between 1000 Hz and 2000 Hz. The **Mel scale** models this:

$$f_{\text{mel}} = 2595 \log_{10} \left(1 + \frac{f_{\text{Hz}}}{700} \right)$$

Mel filterbank: $N_{\text{mels}} = 64$ triangular filters uniformly spaced on the Mel scale between $f_{\text{min}} = 20$ Hz and $f_{\text{max}} = 11,025$ Hz. Each filter sums the power spectrum over a band, giving a perceptually weighted representation.

Why log compression? The dynamic range of acoustic signals is enormous. The log operation $\log(P + \varepsilon)$ compresses this range, making quiet murmurs and loud valve sounds comparably represented in the feature map. This mimics the logarithmic response of the hair cells in the cochlea.

3. Dataset and Preprocessing

3.1 The Five Classes

Label	Full Name	Clinical Description
AS	Aortic Stenosis	Systolic ejection murmur; high-velocity aortic jet
MR	Mitral Regurgitation	Holosystolic murmur; backward flow through mitral valve
MS	Mitral Stenosis	Diastolic rumble; restricted mitral inflow
MVP	Mitral Valve Prolapse	Mid-systolic click; late systolic murmur
N	Normal	S1 and S2 only; no murmur

3.2 Data Pipeline

3.2.1 Step 1: Resampling

Each recording is resampled to $f_s = 22,050$ Hz using `librosa`. This standardizes the frequency axis regardless of the original acquisition rate. The algorithm used is a polyphase filter with a Kaiser window for anti-aliasing.

3.2.2 Step 2: Fixed-Length Windowing

Every signal is either trimmed (if $> L$ samples) or zero-padded on the right (if $< L$) to produce exactly $L = 66,150$ samples ($= 3.0$ s). This produces uniform tensors required for batched GPU computation.

Remark 3.1. Zero-padding adds zeros to the end of the signal, not to the beginning. This preserves the cardiac content (which typically starts early in the recording) and pads with silence. The model must learn to ignore the padding; global average pooling over the time dimension reduces the impact of padding.

3.2.3 Step 3: Amplitude Normalization

$$\tilde{s}[n] = \frac{s[n]}{\max_n |s[n]|}$$

This maps every recording into $[-1, 1]$. Without normalization, differences in microphone gain or patient-to-device distance would dominate the classification signal.

3.3 Train / Validation / Test Split

$$N_{\text{val}} = N_{\text{test}} = \lfloor 0.15 N_{\text{total}} \rfloor, \quad N_{\text{train}} = N_{\text{total}} - N_{\text{val}} - N_{\text{test}}.$$

A fixed random seed (42) ensures reproducibility. The test set is held out entirely during training — it is only accessed once, for final evaluation. Validation loss and accuracy guide early stopping and hyperparameter selection.

4. Neural Operator Theory: From CNNs to FNOs

4.1 What Is an Operator?

A standard neural network maps a finite-dimensional vector to a finite-dimensional vector: $f_\theta : \mathbb{R}^n \rightarrow \mathbb{R}^m$. An **operator** maps a function to a function: $\mathcal{F} : u \mapsto v$ where u, v are functions on some domain Ω .

Example 4.1. The operation “apply the convolution kernel k to input function u ” is an operator: $(\mathcal{K}u)(x) = \int_{\Omega} k(x - y) u(y) dy$.

A **neural operator** is an operator whose parameters are learned from data.

4.2 Convolutional Neural Networks as Local Operators

A 1D CNN with kernel w of width p computes:

$$(\mathcal{W}u)[n] = \sum_{j=-p/2}^{p/2} w[j] u[n - j]$$

This is a **local** operator: the output at position n depends only on a window of p neighboring inputs. For a PCG signal of length $L = 66,150$:

- A kernel of width $p = 3$ samples sees $3/22050 \approx 0.14$ ms of signal.

- To relate S1 to S2 (separated by ~ 300 ms = 6615 samples), we need either very wide kernels (inefficient) or many stacked layers.

Why Local Kernels Struggle with PCG

A murmur’s diagnostic significance depends on *when in the cardiac cycle it occurs* (systolic vs. diastolic). A local CNN kernel sees only a small patch of the signal and cannot, in a single layer, determine where in the cycle that patch falls. Many stacked layers increase the effective receptive field, but at the cost of many parameters and potential gradient instability.

4.3 Fourier Convolution: A Global Operator

The convolution theorem states:

$$(k * u)(x) = \int k(x - y) u(y) dy \iff \widehat{(k * u)}(\xi) = \hat{k}(\xi) \cdot \hat{u}(\xi)$$

Convolution in the time domain is equivalent to **pointwise multiplication in the frequency domain**. This is the core insight exploited by the FNO.

4.3.1 Proof of the Convolution Theorem

Derivation

$$\begin{aligned} \widehat{(k * u)}(\xi) &= \int e^{-2\pi i \xi x} \left(\int k(x - y) u(y) dy \right) dx \\ &= \int u(y) \left(\int k(x - y) e^{-2\pi i \xi x} dx \right) dy \\ &= \int u(y) e^{-2\pi i \xi y} \left(\int k(z) e^{-2\pi i \xi z} dz \right) dy \quad (z = x - y) \\ &= \hat{k}(\xi) \int u(y) e^{-2\pi i \xi y} dy = \hat{k}(\xi) \hat{u}(\xi). \quad \square \end{aligned}$$

4.4 The Fourier Neural Operator (FNO)

Core FNO Idea

Instead of learning a convolution kernel $k(x)$ in the time domain (which has infinitely many parameters for a global operator), learn its Fourier transform $\hat{k}(\xi)$ directly. Truncate to the M most informative modes. This gives a parameter-efficient global operator.

4.4.1 SpectralConv1d: The Core Layer

Given $\mathbf{x} \in \mathbb{R}^{B \times C_{\text{in}} \times L}$ (batch $\times$ channels $\times$ length):

Step 1: Real FFT.

$$\hat{\mathbf{x}} = \text{rfft}(\mathbf{x}) \in \mathbb{C}^{B \times C_{\text{in}} \times (L/2+1)}$$

Step 2: Retain M modes and apply complex linear mixing.

$$\hat{\mathbf{y}}_{b,o,k} = \sum_{i=1}^{C_{\text{in}}} \hat{\mathbf{x}}_{b,i,k} \hat{W}_{i,o,k}, \quad k = 0, \dots, M-1$$

where $\hat{W} = W_r + iW_i$ with learnable real tensors $W_r, W_i \in \mathbb{R}^{C_{\text{in}} \times C_{\text{out}} \times M}$. High-frequency modes $k \geq M$ are set to zero.

Step 3: Inverse FFT.

$$\mathbf{y} = \text{irfft}(\hat{\mathbf{y}}, n = L) \in \mathbb{R}^{B \times C_{\text{out}} \times L}$$

What SpectralConv1d Learns

For each of the M retained Fourier modes, the matrix $\hat{W}_{:, :, k} \in \mathbb{C}^{C_{\text{in}} \times C_{\text{out}}}$ learns how to mix input channels at that specific frequency. Two modes at similar frequencies can be mixed differently. The truncation to M modes is a **low-pass inductive bias**: we assume cardiac sounds are bandlimited (which they are, below ~ 1 kHz).

Why complex weights? A Fourier coefficient $\hat{x}_k = A_k e^{i\phi_k}$ carries both amplitude A_k and phase ϕ_k . A complex weight $\hat{W}_k = R_k e^{i\psi_k}$ can change both the amplitude ($A_k \rightarrow R_k A_k$) and the phase ($\phi_k \rightarrow \phi_k + \psi_k$) of each mode independently.

4.4.2 Initialization and Scaling

Weights are initialized as $W_r, W_i \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 1/\sqrt{C_{\text{in}} \cdot C_{\text{out}}}$. This is a standard Xavier/Glorot-style initialization that keeps the output variance of order 1 regardless of channel count, preventing activation explosion or vanishing at initialization.

4.4.3 FNO Block: Spectral + Local Paths

Each FNOBlock1d combines the spectral convolution with a **bypass path**:

$$\mathbf{x}^{(\ell+1)} = \text{GELU}\left(\text{InstanceNorm}\left(\underbrace{\mathcal{K}_\ell \cdot \mathbf{x}^{(\ell)}}_{\text{spectral: global, frequency-domain}} + \underbrace{W_\ell \mathbf{x}^{(\ell)}}_{\text{bypass: local, } 1 \times 1 \text{ conv}}\right)\right)$$

Why a bypass? The spectral convolution is global but operates in frequency space and cannot model the *phase relationships* between different time positions directly. The 1×1 bypass convolution provides instantaneous channel mixing in the time domain. Together, the two paths give the block both global spectral awareness and local temporal flexibility.

InstanceNorm: Normalizes each channel independently across the time dimension for each sample in the batch:

$$\text{IN}(x_{b,c}) = \gamma_c \frac{x_{b,c} - \mu_{b,c}}{\sigma_{b,c} + \varepsilon} + \beta_c$$

where $\mu_{b,c} = \frac{1}{L} \sum_n x_{b,c,n}$ and $\sigma_{b,c}^2 = \frac{1}{L} \sum_n (x_{b,c,n} - \mu_{b,c})^2$. Unlike BatchNorm, this is computed per sample (not per batch), making it robust to variable batch sizes and variable signal amplitudes.

GELU activation: $\text{GELU}(x) = x \Phi(x)$ where Φ is the Gaussian CDF. GELU is smooth everywhere (unlike ReLU, which has a kink at 0), making gradients more stable. It is the activation function used in BERT, GPT, and ViT.

4.4.4 Overall FNO Backbone

Stage	Module	Input shape	Output shape	Params
Mask	PhysioFrequencyMask	$(B, 1, L)$	$(B, 1, L)$	$M + 4$
Lift	Conv1d(1,W,1)	$(B, 1, L)$	(B, W, L)	$W + W$
Block 1	FNOBlock1d	(B, W, L)	(B, W, L)	see below
Block 2	FNOBlock1d	(B, W, L)	(B, W, L)	
Block 3	FNOBlock1d	(B, W, L)	(B, W, L)	
Block 4	FNOBlock1d	(B, W, L)	(B, W, L)	
Pool	<code>x.mean(dim=-1)</code>	(B, W, L)	(B, W)	0
MLP	Linear $\times 2$	(B, W)	$(B, 64)$	

With $W = 64$, $M = 20$, $D = 4$ FNO blocks. Each FNO block has $2 \times C_{\text{in}} \times C_{\text{out}} \times M$ (spectral weights) + C^2 (bypass) + $2C$ (InstanceNorm affine) parameters.

Global average pooling: $\bar{\mathbf{x}} = \frac{1}{L} \sum_{n=1}^L \mathbf{x}_{:,n}$. This is length-invariant, making the classifier robust to zero-padding variations.

MLP projection head: Linear($W, 128$) $\rightarrow$ GELU $\rightarrow$ Dropout(0.3) $\rightarrow$ Linear(128, 64) $\rightarrow$ GELU produces the 64-dim embedding $\mathbf{z}$.

5. PhysioFrequencyMask: Learned Spectral Gating

5.1 Motivation

Before the signal enters the FNO backbone, we want to suppress irrelevant frequencies and amplify physiologically relevant ones. Rather than using a fixed filter (which cannot adapt), we use a learnable mask that is *initialized from physiological knowledge* and *adapts to data*.

5.2 Component 1: Per-Mode Free Logit

5.2.1 Initialization

We know from physiology that S1 energy is in 25–45 Hz and S2 energy in 50–70 Hz. We initialize a mask $m_k^{(0)}$ for each of the $M = 20$ retained modes:

$$m_k^{(0)} = \begin{cases} 1.0 & f_k \in [25, 45] \text{ Hz or } f_k \in [50, 70] \text{ Hz} \\ 0.1 & \text{otherwise} \end{cases}$$

5.2.2 Logit Parameterization

If we learn m_k directly, gradient descent might push it below 0 or above 1. To keep $m_k \in (0, 1)$ without projection, we parameterize in *logit space*:

$$\theta_k = \ln \left(\frac{m_k^{(0)}}{1 - m_k^{(0)}} \right), \quad m_k = \text{sigmoid}(\theta_k)$$

The logit θ_k is unconstrained ($\theta_k \in \mathbb{R}$), but $\text{sigmoid}(\theta_k) \in (0, 1)$ always. For $m_k^{(0)} = 1.0$: $\theta_k = +\infty$ (approximately $+13.8$ after clipping to $1 - 10^{-6}$). For $m_k^{(0)} = 0.1$: $\theta_k = \ln(0.1/0.9) \approx -2.20$.

5.2.3 Forward Pass

The per-mode mask is extended to the full rfft length, with non-retained modes set to 0.1:

$$\text{fm}[k] = \begin{cases} \text{sigmoid}(\theta_k) & k < M \\ 0.1 & k \geq M \end{cases}$$

Applied in the Fourier domain:

$$\tilde{x}_{\text{masked}}(t) = \text{irfft}(\hat{x}(\xi) \cdot \text{fm}(\xi), n = L)$$

5.3 Component 2: Learnable Band Boundaries

5.3.1 Motivation

The per-mode logit can learn any mask shape, but it treats each mode independently. We also want to learn *where the S1 and S2 bands sit*, as a smooth interpretable structure. Four scalar parameters encode this: $s1_lo, s1_hi, s2_lo, s2_hi \in \mathbb{R}$.

5.3.2 Sigmoid Parameterization of Band Boundaries

The boundaries must lie in $[0, f_s/2]$. We store them as raw reals and map via:

$$f_{S1}^{lo} = \text{sigmoid}(s1_lo) \cdot \frac{f_s}{2}, \quad f_{S1}^{hi} = \text{sigmoid}(s1_hi) \cdot \frac{f_s}{2}, \quad \text{etc.}$$

Initialized from $s1_lo = \text{logit}(25/11025)$, etc.

5.3.3 Double-Sigmoid Bump Function

A differentiable soft indicator for a band (f_{lo}, f_{hi}) :

$$B(f; f_{lo}, f_{hi}) = \text{sigmoid}(k(f - f_{lo})) \cdot \text{sigmoid}(k(f_{hi} - f))$$

with sharpness $k = 50$. This function is:

- ≈ 0 for $f \ll f_{lo}$ or $f \gg f_{hi}$ (outside the band)
- ≈ 1 for $f_{lo} \ll f \ll f_{hi}$ (inside the band)
- Smooth and differentiable everywhere (gradients can flow through f_{lo}, f_{hi})

5.3.4 Combined Mask

$$\text{fm}[k] = \text{clip}(\text{sigmoid}(\theta_k) + \alpha(B_{S1}(f_k) + B_{S2}(f_k)), 0, 1)$$

with bump weight $\alpha = 0.3$. The two components are complementary:

- **Per-mode logits θ_k** : maximum flexibility, can learn any shape.

- **Band bumps:** constrained to be band-shaped, but *the band location and width are themselves learnable*. After training, calling `get_bands()` reveals where the network decided S1 and S2 actually sit.

Remark 5.1. The `PhysioFrequencyMask` has $M + 4$ learnable parameters: $M = 20$ per-mode logits and 4 band boundary scalars. This is an extremely small number relative to the total model size, making the mask’s contribution highly interpretable.

6. PhysioFeatureExtractor: 18 Explicit Physical Quantities

6.1 The Structural Physics Principle

Structural vs. Penalized Physics

Penalized approach (old): $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_1 \mathcal{L}_{\text{periodicity}} + \lambda_2 \mathcal{L}_{\text{hierarchy}}$. The penalty terms are computed on the fixed raw input signal. Since no network parameters appear inside them, $\partial \mathcal{L}_{\text{periodicity}} / \partial \theta = 0$ for every model parameter θ . They contribute zero gradient. They affect the printed loss number but not the optimization.

Structural approach (current): Physical quantities are computed and provided as direct inputs to the classifier. Cross-entropy $\mathcal{L}_{\text{CE}}$ then supervises the classifier to use them. Gradients flow through the feature computations into the learnable band parameters.

6.2 Full Feature Table

#	Feature	Physical meaning	Learnable?
1	S1 spectral energy	Power in S1 valve closure band	S1 boundaries
2	S2 spectral energy	Power in S2 valve closure band	S2 boundaries
3	S1 spectral centroid	Freq. center of mass, S1 band	S1 boundaries
4	S2 spectral centroid	Freq. center of mass, S2 band	S2 boundaries
5	Murmur bandwidth	Spectral spread of turbulent flow region	Murmur boundaries
6	Harmonic energy f_0	Energy at fundamental HR	HR band + h_w
7	Harmonic energy $2f_0$	Energy at 2nd harmonic	HR band + h_w
8	Harmonic energy $3f_0$	Energy at 3rd harmonic	HR band + h_w
9	Harmonic energy $4f_0$	Energy at 4th harmonic	HR band + h_w
10	Dominant frequency	Heart rate estimate	HR search band
11	Periodicity ratio	Fraction of energy in cardiac band	Fixed
12	RMS amplitude	Overall signal energy	Fixed
13	S1/S2 energy ratio	Relative valve closure intensity (log)	S1/S2 boundaries
14	Envelope mean	Mean instantaneous amplitude	Fixed
15	Envelope std	Variability of instantaneous amplitude	Fixed
16	Inst. freq. mean	Mean phase derivative of analytic signal	Fixed
17	S1-to-S2 interval	Systolic duration proxy ($1/2f_0$)	HR band
18	Cross-beat consistency	Harmonic energy ratio over cycles	HR band

6.3 Learnable Parameters in PhysioFeatureExtractor

Parameter	Shape	Init	Controls
s1_lo, s1_hi	scalar $\times 2$	25, 45 Hz	S1 band boundaries
s2_lo, s2_hi	scalar $\times 2$	50, 70 Hz	S2 band boundaries
mu_lo, mu_hi	scalar $\times 2$	100, 500 Hz	Murmur band boundaries
hr_lo, hr_hi	scalar $\times 2$	0.8, 3.0 Hz	Heart rate search range
harmonic_w	(4,)	0.25 each	Weights on f_0 – $4f_0$

Total: $8 + 4 = 12$ learnable parameters in the extractor. All band boundaries use the sigmoid parameterization. `harmonic_w` is passed through softmax before use, so the 4 weights always sum to 1.

6.4 Soft Band Energy

Differentiable Band Energy

For a band (f_{lo}, f_{hi}) with soft indicator $B(f; f_{lo}, f_{hi})$:

$$E_{S1} = \sum_{k=0}^{L/2} P[k] \cdot B(f_k; f_{S1}^{lo}, f_{S1}^{hi})$$

where $P[k] = |\hat{s}[k]|^2$ is the PSD. This is a smooth function of f_{S1}^{lo} and f_{S1}^{hi} :

$$\frac{\partial E_{S1}}{\partial f_{S1}^{lo}} = \sum_k P[k] \frac{\partial B(f_k; f_{S1}^{lo}, f_{S1}^{hi})}{\partial f_{S1}^{lo}} = \sum_k P[k] \frac{-k \cdot \text{sigmoid}'(k(f_k - f_{S1}^{lo})) \cdot \text{sigmoid}(k(f_{S1}^{hi} - f_k))}{1}$$

The gradient concentrates at the band edge $f_k \approx f_{S1}^{lo}$, which is where the boundary actually is. This allows gradient descent to move the boundary to wherever the data says the S1 band should be.

6.5 Spectral Centroid

The frequency center of mass within a band:

$$\bar{f}_{S1} = \frac{\sum_k P[k] \cdot B(f_k; f_{S1}^{lo}, f_{S1}^{hi}) \cdot f_k}{\sum_k P[k] \cdot B(f_k; f_{S1}^{lo}, f_{S1}^{hi}) + \varepsilon}$$

Clinical interpretation: In aortic stenosis, the turbulent murmur shifts spectral energy toward higher frequencies, so the spectral centroid of the S1 region may be higher than in a normal recording, even if the S1 sound itself is partially masked by the murmur.

6.6 Murmur Bandwidth

The spectral *spread* in the murmur band $(f_{\mu}^{lo}, f_{\mu}^{hi})$:

$$\sigma_{\mu}^2 = \frac{\sum_k P[k] \cdot B_{\mu}(f_k) \cdot (f_k - \bar{f}_{\mu})^2}{\sum_k P[k] \cdot B_{\mu}(f_k) + \varepsilon}, \quad \text{BW}_{\mu} = \sqrt{\sigma_{\mu}^2}$$

Clinical interpretation: A broad murmur (high bandwidth) in the 100–500 Hz range is characteristic of turbulent flow in AS. A narrower peak might indicate a more specific resonance.

6.7 Harmonic Analysis and the Heart Rate

The cardiac cycle is quasi-periodic at the heart rate f_0 . In the frequency domain, this periodicity manifests as peaks at $f_0, 2f_0, 3f_0, 4f_0, \dots$ (the fundamental and its harmonics).

6.7.1 Finding the Dominant Frequency

We search for the peak of the PSD within the heart rate band ($f_{\text{HR}}^{\text{lo}}, f_{\text{HR}}^{\text{hi}}$) using the soft indicator:

$$f_0 = f_{k^*}, \quad k^* = \arg \max_k [P[k] \cdot B_{\text{HR}}(f_k)]$$

The `argmax` is not differentiable. This is acceptable because f_0 is used to compute energies at fixed multiples, and the gradients flow through the energy computation, not through the `argmax` itself.

6.7.2 Harmonic Energies

For each harmonic $n \in \{1, 2, 3, 4\}$:

$$E_{nf_0} = \sum_k P[k] \cdot B(f_k; nf_0 - \delta, nf_0 + \delta), \quad \delta = 2 \text{ Hz}$$

The weighted sum with learnable softmax weights:

$$w = \text{softmax}(\text{harmonic_w}) \in \mathbb{R}^4, \quad \sum_{n=1}^4 w_n = 1$$

allows the model to learn which harmonic is most discriminative.

Why harmonics matter clinically? In normal hearts, most energy is at f_0 and falls off rapidly at higher harmonics. In pathological conditions, the harmonic structure changes:

- AS: additional energy at non-harmonic frequencies due to the turbulent murmur.
- MS: weaker fundamental because diastolic filling is impaired.
- MVP: the click adds a transient that spreads energy across harmonics.

6.8 Periodicity Ratio

$$r_{\text{per}} = \frac{\sum_{k: f_k \in [1.0, 5.3] \text{ Hz}} P[k]}{\sum_k P[k] + \varepsilon}$$

This measures the fraction of total signal energy concentrated in the cardiac HR band and its first 4 harmonics (covering 60–80 BPM). A value near 1 indicates a highly periodic signal; a value near 0 indicates a noisy or aperiodic signal.

Former role: This was previously computed as a loss penalty $\mathcal{L}_{\text{periodicity}}$ on the raw input. Since raw input has no model parameters, that gave zero gradient. Now it is feature 11: a number given directly to the classifier.

6.9 RMS Amplitude

$$r_{\text{rms}} = \sqrt{\frac{1}{L} \sum_{n=0}^{L-1} s[n]^2}$$

After amplitude normalization, r_{rms} reflects the *shape* of the signal (an impulsive click has lower RMS than a sustained murmur at the same peak amplitude).

6.10 S1/S2 Energy Ratio

$$\rho = \frac{E_{S1}}{E_{S2} + \varepsilon}$$

Provided to the classifier as $\log(1 + \rho)$ (log scale compresses the dynamic range). This encodes the physiology that S1 is normally louder than S2. In pathological conditions, this ratio can change:

- AS: systolic murmur overlaps with S1 band, potentially inflating E_{S1} .
- MS: diastolic murmur may elevate E_{S2} , reducing ρ .

Former role: This was enforced as $\mathcal{L}_{\text{hierarchy}} = \text{ReLU}(E_{S2} - E_{S1})$. Again, computed on fixed input, zero gradient. Now it is feature 13.

6.11 The Hilbert Transform and Analytic Signal

6.11.1 The Analytic Signal

For a real signal $s(t)$, the **analytic signal** is:

$$s_a(t) = s(t) + i\mathcal{H}[s](t)$$

where $\mathcal{H}[s]$ is the Hilbert transform. The analytic signal has:

- **Envelope:** $e(t) = |s_a(t)| = \sqrt{s(t)^2 + (\mathcal{H}[s])^2}$
- **Instantaneous phase:** $\varphi(t) = \arg(s_a(t)) = \arctan\left(\frac{\mathcal{H}[s](t)}{s(t)}\right)$
- **Instantaneous frequency:** $\text{IF}(t) = \frac{1}{2\pi} \frac{d\varphi}{dt}$

6.11.2 Computing the Hilbert Transform via FFT

The Hilbert transform doubles the positive-frequency components and zeros the negative:

$$\hat{h}(\xi) = \begin{cases} -i & \xi > 0 \\ 0 & \xi = 0 \\ i & \xi < 0 \end{cases}$$

In terms of the rfft (which only stores non-negative frequencies):

$$S(\xi) = \text{rfft}(s), \quad \tilde{S}[k] = h[k] \cdot S[k], \quad \mathcal{H}[s] = \text{irfft}(\tilde{S})$$

where $h[0] = 1$, $h[k] = 2$ for $0 < k < L/2$, $h[L/2] = 1$ (even L).

Why This Formula Works

For the rfft, we only have the non-negative frequency components. The Hilbert transform for real signals is equivalent to: (1) keep the DC and Nyquist components as-is, (2) multiply all positive-frequency components by $-i$ (or equivalently, double them before the irfft, which gives the imaginary part of the analytic signal). The factor of 2 accounts for the energy in the negative frequencies that are discarded.

6.11.3 Envelope Mean and Std

The envelope $e(t)$ captures the *instantaneous amplitude* of the signal. For a PCG:

- At S1 onset, $e(t)$ spikes upward.
- During silence between S1 and S2, $e(t)$ is near zero.
- A murmur produces sustained elevated $e(t)$ throughout systole.

$\text{env_mean} = \frac{1}{L} \sum_n e[n]$ and $\text{env_std} = \sqrt{\frac{1}{L} \sum_n (e[n] - \text{env_mean})^2}$ together capture the average loudness and how much the loudness varies over the recording.

6.11.4 Instantaneous Frequency Mean

$$\text{IF}[n] = \frac{\varphi[n+1] - \varphi[n]}{2\pi} \cdot f_s, \quad \overline{\text{IF}} = \frac{1}{L-1} \sum_{n=0}^{L-2} \text{IF}[n]$$

The mean instantaneous frequency is a time-domain measure of the *average oscillation rate* of the signal. In normal recordings, $\overline{\text{IF}}$ reflects the dominant heart rate. In murmur recordings, the IF is elevated because of the higher-frequency turbulent components.

6.12 S1-to-S2 Interval Estimate

$$\Delta_{S1S2} = \frac{1}{2f_0}$$

The dominant frequency f_0 (heart rate) gives the period $T = 1/f_0$. Systole occupies approximately half the cycle, so $\Delta_{S1S2} \approx T/2$. This is a rough proxy but provides the classifier with an explicit estimate of the systolic duration.

Clinical value: In aortic stenosis, systole is prolonged (delayed closure of the stiffened aortic valve), so Δ_{S1S2} is expected to be longer than normal.

6.13 Cross-Beat Consistency

$$C = \frac{\sum_{n=1}^4 \sum_k P[k] \cdot B(f_k; nf_0 - \delta, nf_0 + \delta)}{\sum_k P[k] + \varepsilon}$$

This measures what fraction of total spectral energy is concentrated at the fundamental heart rate and its first three harmonics. A perfectly periodic cardiac signal would have $C \approx 1$. A signal with arrhythmia, noise, or pathological transients would have $C \ll 1$.

Vectorized implementation: Instead of looping over samples (which breaks gradient flow), we broadcast: $nf_0 \in \mathbb{R}^B$ becomes $(nf_0)^{\text{unsqueeze}(1)} \in \mathbb{R}^{B \times 1}$, and $f \in \mathbb{R}^{L/2+1}$ becomes $f^{\text{unsqueeze}(0)} \in \mathbb{R}^{1 \times L/2+1}$. The band mask $(B_n) \in \mathbb{R}^{B \times L/2+1}$ is then computed in parallel for all samples.

7. Classification Architecture: Fusing FNO and Physics

7.1 The Two-Path Architecture

The full `FNOClassifier` has two parallel processing paths:

$$\begin{aligned}
 \text{Path 1 (FNO)} : x &\xrightarrow{\text{PhysioMask}} x_{\text{filt}} \xrightarrow{\text{Lift}} \xrightarrow{4 \times \text{FNOBlock}} \xrightarrow{\text{GAP}} \xrightarrow{\text{MLP}} \mathbf{z} \in \mathbb{R}^{64} \\
 \text{Path 2 (Physics)} : x &\xrightarrow{\text{PhysioFeatureExtractor}} \mathbf{p} \in \mathbb{R}^{18} \\
 \text{Fusion} : [\mathbf{z}; \mathbf{p}] &\in \mathbb{R}^{82} \xrightarrow{\text{Linear}(82,5)} \hat{y} \in \mathbb{R}^5
 \end{aligned}$$

Why concatenate rather than add? Concatenation allows the classifier to learn the relative importance of the FNO embedding vs. the physics features independently. If the physics features are highly discriminative (e.g., the S1/S2 ratio clearly separates MS from Normal), the linear classifier can weight them heavily. If the FNO embedding captures something the physics features miss (e.g., fine spectral texture in the murmur), it will be weighted accordingly.

7.2 The Independent Band Parameters

A critical architectural decision: `PhysioFrequencyMask` and `PhysioFeatureExtractor` each have their own `s1_lo`, `s1_hi`, `s2_lo`, `s2_hi` parameters, starting at the same values but updated independently.

- **Mask boundaries:** Optimize for what to pass through to the FNO. The FNO backbone processes the filtered signal, so the mask boundaries are optimized for the downstream spectral features.
- **Extractor boundaries:** Optimize for the most discriminative physical feature values. A wider S1 band in the extractor might give a more stable energy estimate; a narrower band in the mask might sharpen the FNO’s spectral focus.

After training, comparing `model.freq_mask.get_bands()` with `model.phys.get_bands()` reveals this divergence — one of the most interpretable outputs of the model.

8. Architecture II: CNN Baseline (Mel Spectrogram)

8.1 Role of the Baseline

The CNN baseline is a **strong conventional architecture** without physiological priors. Comparing it to the FNO+Extractor answers: *does our physics-informed design actually improve performance, or is a generic architecture sufficient?*

A positive accuracy gap ($\text{FNO} > \text{CNN}$) provides empirical justification for the architectural complexity. A zero gap would suggest the physics priors are not helping and the design should be reconsidered.

8.2 MelSpectrogramLayer: A Differentiable Front-End

8.2.1 Why a Differentiable Spectrogram?

At inference time, we need only PyTorch — no librosa dependency. The `MelSpectrogramLayer` implements the entire spectrogram pipeline as a fixed (non-learnable) differentiable transform using `register_buffer`.

8.2.2 Step 1: STFT

$$X(m, k) = \sum_{n=0}^{N-1} s[n + mh] \cdot w[n] \cdot e^{-2\pi i kn/N}, \quad N = 512, h = 128$$

Output shape: $(B, N/2 + 1, T)$ complex, where $T = \lfloor (L - N)/h \rfloor + 1$ is the number of time frames.

8.2.3 Step 2: Mel Filterbank Application

The filterbank matrix $\mathbf{F} \in \mathbb{R}^{N_{\text{mels}} \times (N/2+1)}$ ($N_{\text{mels}} = 64$) maps each power spectrum frame to Mel bins:

$$P_{\text{mel}}(m) = \mathbf{F} \cdot |X(m, \cdot)|^2 \in \mathbb{R}^{N_{\text{mels}}}$$

8.2.4 Step 3: Log Compression

$$S_{\text{mel}}(m, k) = \log(P_{\text{mel}}(m, k) + 10^{-6})$$

Why $\varepsilon = 10^{-6}$? Without it, $\log(0) = -\infty$ for silent frames. ε is the minimum representable energy — since the signal is in $[-1, 1]$, $\varepsilon = 10^{-6}$ is well below any genuine signal energy.

8.3 CNN Backbone

Three 2D convolutional blocks process $S_{\text{mel}} \in \mathbb{R}^{B \times 1 \times N_{\text{mels}} \times T}$:

Block	Channels	Operation	Output
1	32	Conv2d(1,32,3,pad=1)+BN+ReLU+MaxPool(2)	$\mathbb{R}^{B \times 32 \times 32 \times T/2}$
2	64	Conv2d(32,64,3,pad=1)+BN+ReLU+MaxPool(2)	$\mathbb{R}^{B \times 64 \times 16 \times T/4}$
3	128	Conv2d(64,128,3,pad=1)+BN+ReLU+AdaptAvgPool(4,4)	$\mathbb{R}^{B \times 128 \times 4 \times 4}$

BatchNorm vs. InstanceNorm: The CNN uses BatchNorm because it processes 2D spectrograms with consistent statistics across the batch. The FNO uses InstanceNorm because it processes 1D signals where the statistics can vary dramatically between recordings.

MLP head: Flatten $\rightarrow$ Linear($128 \times 16, 256$) $\rightarrow$ ReLU $\rightarrow$ Dropout(0.4) $\rightarrow$ Linear(256, 5).

8.4 Architecture Comparison

Aspect	FNO + Extractor	CNN Baseline
Input domain	Raw waveform (1-D)	Log-Mel spectrogram (2-D)
Physics	Structural (mask + 18 features)	None
Long-range deps.	Global (full-signal FFT)	Local (3×3 kernel)
Classifier input	82-dim (64 FNO + 18 phys)	Fixed CNN features
Learnable physics	12 band/harmonic params	0
Training loss	$\mathcal{L}_{\text{CE}}$ only	$\mathcal{L}_{\text{CE}}$ only
Interpretability	Mask, bands, harmonics	Limited

9. Loss Function: Pure Cross-Entropy

9.1 The Single Loss

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(\hat{y}, y) = -\frac{1}{B} \sum_{b=1}^B \log \left(\frac{\exp(\hat{y}_{b,y_b})}{\sum_{c=0}^4 \exp(\hat{y}_{b,c})} \right)$$

9.2 Why No Auxiliary Loss Terms?

The original design used:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_1 \mathcal{L}_{\text{per}} + \lambda_2 \mathcal{L}_{\text{hier}}$$

The problem: Both $\mathcal{L}_{\text{per}}$ and $\mathcal{L}_{\text{hier}}$ were computed as:

$$\mathcal{L}_{\text{per}} = 1 - \frac{\sum_{f \in \mathcal{B}_{\text{HR}}} P_b(f)}{\sum_f P_b(f) + \varepsilon} \quad \text{where } P_b = |\text{rfft}(x_b)|^2, \quad x_b \text{ is fixed input data}$$

$$\mathcal{L}_{\text{hier}} = \frac{1}{B} \sum_b \text{ReLU}(E_{S2}^{(b)} - E_{S1}^{(b)}) \quad \text{where } E_{S1}, E_{S2} \text{ are computed from } x_b$$

x_b is the raw input batch. For any model parameter θ : $\partial x_b / \partial \theta = 0$ because x_b is data, not a network output. Therefore $\partial \mathcal{L}_{\text{per}} / \partial \theta = 0$ and $\partial \mathcal{L}_{\text{hier}} / \partial \theta = 0$.

The constraints were printed as part of the total loss but never influenced the optimization.

The structural fix: Periodicity (feature 11) and S1/S2 ratio (feature 13) are now computed inside `PhysioFeatureExtractor` and passed to the classifier. The classifier learns to *use* them via $\mathcal{L}_{\text{CE}}$. If periodicity is discriminative, the corresponding weight in the linear classifier grows. If it is not, the weight stays near zero.

9.3 Cross-Entropy: Mathematical Properties

9.3.1 Equivalence to Maximum Likelihood

Cross-entropy is the negative log-likelihood of the correct label under the model's predicted categorical distribution $p_c = \exp(\hat{y}_c) / \sum_{c'} \exp(\hat{y}_{c'})$:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{B} \sum_b \log p_{y_b} = -\frac{1}{B} \sum_b \log q_\theta(y_b|x_b)$$

Minimizing $\mathcal{L}_{\text{CE}}$ is equivalent to maximizing the log-likelihood of the data under the model.

9.3.2 Why Not Mean Squared Error?

For classification, MSE loss $\frac{1}{B} \sum_b \|\hat{y}_b - \mathbf{1}_{y_b}\|^2$ has a gradient that is proportional to $(\hat{y} - \mathbf{1}_y)$, which saturates for very confident (but wrong) predictions, producing vanishing gradients. Cross-entropy has gradient $\propto (p_c - \mathbf{1}_{c=y})$, which is large when the model is confidently wrong, providing strong corrective gradient.

10. Training Procedure

10.1 AdamW Optimizer

Adam and AdamW

Adam (Adaptive Moment Estimation) maintains per-parameter adaptive learning rates based on running estimates of the first moment (mean) and second moment (variance) of the gradients:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, & v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \\ \hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, & \hat{v}_t &= \frac{v_t}{1 - \beta_2^t} \quad (\text{bias correction}) \\ \theta_{t+1} &= \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon} \end{aligned}$$

AdamW decouples weight decay from the gradient update:

$$\theta_{t+1} = \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon} - \alpha \lambda_{\text{wd}} \theta_t$$

In Adam, L2 regularization adds $\lambda \theta$ to the gradient before computing $\hat{m}_t$ and $\hat{v}_t$, which means the effective weight decay is scaled by $1/\sqrt{\hat{v}_t}$ — different for each parameter. AdamW applies weight decay directly to the parameter, independent of the gradient statistics. This gives better regularization properties, especially for transformers and FNOs.

We use $\alpha = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, $\lambda_{\text{wd}} = 10^{-4}$.

10.2 Cosine Annealing Learning Rate Schedule

$$\alpha_t = \alpha_{\min} + \frac{\alpha_{\max} - \alpha_{\min}}{2} \left(1 + \cos\left(\frac{\pi t}{T}\right) \right), \quad T = 60$$

with $\alpha_{\min} = 0$, $\alpha_{\max} = 10^{-3}$.

Why cosine annealing?

- **Early training:** Large α allows the model to make rapid progress. The optimizer can escape local minima.
- **Late training:** Small α allows fine-grained convergence to the minimum. The optimizer stops oscillating around the minimum.
- **Smooth transition:** The cosine curve is smooth, avoiding abrupt changes in gradient statistics that could destabilize training.

10.3 Gradient Clipping

Before each optimizer step, the global ℓ^2 norm of all gradients is computed:

$$\|g\|_2 = \sqrt{\sum_i \|g_i\|_F^2}$$

If $\|g\|_2 > 1.0$, all gradients are rescaled: $g_i \leftarrow g_i \cdot 1.0/\|g\|_2$.

Why is clipping necessary for FNOs? The FFT is a linear transform, but spectral operations can produce large values when Fourier modes are large. A single unusually loud recording in a batch could produce a gradient spike that destroys the parameter values learned from many other recordings. Clipping at norm 1.0 bounds the worst-case update magnitude.

10.4 Model Checkpointing

At the end of each epoch, if the validation accuracy exceeds the best seen so far, the model state dict is saved to disk. After training completes, the best checkpoint is restored. This prevents selecting an overfitted model from the final epoch.

11. Evaluation: Learned Physical Quantities

11.1 What We Can Learn from the Trained Parameters

After training, every learnable parameter carries information about the cardiac acoustics in the dataset. Reading out these parameters is a form of **physics discovery**: the model has been given physical priors and has adapted them to the data.

11.2 The Frequency Mask

`model.freq_mask.get_mask()` returns $\{m_k\}_{k=0}^{M-1}$. Plotted against $\{f_k\}$, this reveals:

1. **S1/S2 weights remain high:** The physiological prior was correct.
2. **Non-S1/S2 weights grew:** Those frequencies are discriminative. This often means the murmur band (100–400 Hz) has grown for AS/MR recordings.
3. **S1/S2 weights dropped:** Surprising — would suggest the band prior is wrong, warranting investigation of data quality.

11.3 Band Boundary Drift

Comparing the initialization and learned values of all band boundaries:

Parameter	Module	Init	Trained	Interpretation
$s1_{lo}, s1_{hi}$	Mask	25, 45 Hz	f_{lo}^*, f_{hi}^*	S1 band for spectral gating
$s2_{lo}, s2_{hi}$	Mask	50, 70 Hz	f_{lo}^{**}, f_{hi}^{**}	S2 band for gating
$s1_{lo}, s1_{hi}$	Extractor	25, 45 Hz	g_{lo}^*, g_{hi}^*	S1 band for feature computation
$s2_{lo}, s2_{hi}$	Extractor	50, 70 Hz	g_{lo}^{**}, g_{hi}^{**}	S2 band for feature
mu_{lo}, mu_{hi}	Extractor	100, 500 Hz		Murmur band
hr_{lo}, hr_{hi}	Extractor	0.8, 3.0 Hz		HR search band

The divergence between mask and extractor S1/S2 boundaries is a unique finding: the mask optimizes for FNO input; the extractor optimizes for feature quality.

11.4 Harmonic Weights

$$w = \text{softmax}(\text{harmonic_w}) \in \mathbb{R}^4, \quad w = [w_1, w_2, w_3, w_4]$$

At initialization, all weights are equal ($w_i = 0.25$). After training:

- If $w_1 \gg w_{2,3,4}$: The fundamental heart rate is most discriminative. This makes sense if the five classes have different HR distributions.
- If $w_{2,3}$ are significant: Higher harmonics carry discriminative information, possibly related to the sharpness of the cardiac cycle (normal vs. pathological).
- Uniform w : All harmonics contribute equally; the model uses the full harmonic structure.

12. Evaluation: t-SNE of FNO Latent Embeddings

12.1 What Is the Latent Embedding?

The FNO path produces a 64-dim vector $\mathbf{z}$ for each recording — the output of the projection MLP before fusion with the physics features. This is the FNO’s learned compressed representation of the cardiac signal.

Remark 12.1. The t-SNE visualizes $\mathbf{z} \in \mathbb{R}^{64}$ (FNO embedding only), not the full $[\mathbf{z}; \mathbf{p}] \in \mathbb{R}^{82}$ (fused representation). This lets us assess how well the FNO alone separates the classes, independent of the explicit physics features.

12.2 t-SNE Algorithm

t-SNE (van der Maaten & Hinton, 2008) reduces high-dimensional embeddings to 2D while approximately preserving *local neighborhood structure*:

Step 1: Define similarities in high-D space via a Gaussian:

$$p_{j|i} = \frac{\exp(-\|z_i - z_j\|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\|z_i - z_k\|^2 / 2\sigma_i^2)}$$

The bandwidth σ_i is set so that the effective number of neighbors equals the **perplexity** parameter (we use 15).

Step 2: Define similarities in 2D space via a heavier-tailed Student t :

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}$$

Step 3: Minimize $\text{KL}(P\|Q) = \sum_{ij} p_{ij} \log(p_{ij}/q_{ij})$ via gradient descent in 2D.

The heavier tail of the Student t in step 2 prevents the *crowding problem*: in high dimensions, most points are approximately equidistant, which would collapse all mapped points to the center. The t distribution allows similar points to be nearby while dissimilar points can be placed far apart without running out of space.

12.3 Interpreting the t-SNE Plot

- **Well-separated clusters:** The FNO has learned class-discriminative representations. Classification by the linear head should be easy.
- **Overlap between MS and MVP:** Clinically expected — both involve diastolic abnormalities and can present with similar murmurs.
- **Normal separated from all pathologies:** The most clinically important separation. If Normal is clearly distinct, the model can screen for any valvular disease without knowing which specific condition is present.
- **Axes have no physical meaning:** The t-SNE axes are not frequency, amplitude, or any physical quantity. They are a 2D projection that maximally preserves neighborhood structure.

13. Confusion Matrices and Classification Metrics

13.1 Confusion Matrix

The normalized confusion matrix $C \in [0, 1]^{5 \times 5}$:

$$C_{ij} = \frac{\text{Number of samples with true class } i \text{ predicted as class } j}{\text{Number of samples with true class } i}$$

Diagonal entries C_{ii} are the **per-class recall** (sensitivity): what fraction of class i samples were correctly identified?

Off-diagonal entries C_{ij} ($i \neq j$) are **confusion rates**: how often was class i mistaken for class j ?

Clinical asymmetry: False negatives ($C_{i,\text{Normal}}$ for $i \neq \text{Normal}$) — pathology predicted as Normal — are more dangerous than false positives ($C_{\text{Normal},j}$ for $j \neq \text{Normal}$) — Normal predicted as pathology. The former might lead to a missed diagnosis; the latter leads to an unnecessary referral.

13.2 Per-Class Metrics

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c}, \quad F1_c = \frac{2 \cdot \text{Prec}_c \cdot \text{Rec}_c}{\text{Prec}_c + \text{Rec}_c}$$

Macro-averaged F1: $\frac{1}{5} \sum_c F1_c$. Treats all classes equally, regardless of class size. Most appropriate for balanced or near-balanced datasets.

Weighted F1: $\sum_c w_c F1_c$ where $w_c = N_c / N_{\text{total}}$. More appropriate for imbalanced datasets, as it weights each class by its prevalence.

14. Summary of Design Choices and Rationale

Choice	Implementation	Rationale
Fourier conv.	spectral <code>SpectralConv1d</code>	PCG is periodic; FFT is the natural basis for periodic signals
Mode truncation $M = 20$	Retain low-freq modes	Cardiac content < 1 kHz; high modes are noise
Per-mode logit	$\theta_k \rightarrow m_k = \text{sigmoid}(\theta_k)$	Unconstrained param, constrained output: $m_k \in (0, 1)$ always
S1/S2-init mask	$m_k^{(0)} = 1.0$ in bands	Physics-guided initialization, not locked; data can override
Learnable bounds	band 4 scalars per module	Boundaries adapt from physiological prior; interpretable post-training
Double-sigmoid bump	$\text{sigmoid}(k(f - f_{lo})) \cdot \text{sigmoid}(k(f_{hi} - f))$	Differentiable; gradients flow through boundary params
Independent params	Mask $\neq$ Extractor	Two distinct optimization objectives; divergence is informative
PhysioFeatureExtractor	18 physical features	Structural physics: explicit inputs, zero-gradient problem solved
Hilbert envelope	FFT-based, differentiable	No external library; gradients flow; captures amplitude modulation
Instantaneous freq.	Phase derivative of analytic	Time-domain frequency estimate; sensitive to murmur activity
Harmonic weights	Softmax over f_0-4f_0	Learnable; reveals which harmonics are discriminative
Fused classifier	<code>Linear(82, 5)</code>	Combines learned FNO features with explicit physics
$\mathcal{L}_{CE}$ only	No aux. penalties	Physics is structural; penalties on fixed input give zero gradient
InstanceNorm in FNO	Per-sample, per-channel	Robust to amplitude variability; appropriate for time series
BatchNorm in CNN	Per-batch, per-channel	Standard for 2D feature maps; stable with fixed spectrogram input
Global avg. pool	<code>x.mean(dim=-1)</code>	Length-invariant; robust to zero-padding
AdamW	Adam + decoupled WD	Better regularization than Adam for spectral models
Cosine annealing	$T = 60$ epochs	Large steps early; fine-grained convergence late
Grad. clipping	$\ g\ _2 \leq 1.0$	Prevents spectral explosion from large Fourier coefficients

15. Connection to Physics-Informed Machine Learning and Continuum Mechanics

15.1 The Physics-Informed Philosophy

The classical approach to machine learning for physical systems is: *collect data, train a model, hope the physics emerges*. The physics-informed approach is: *encode what you know about the physics into the architecture, so the model only needs to learn what the data tells it above and beyond the physics*.

15.2 Analogies with Continuum Mechanics

Five Direct Analogies

1. Fourier space as natural basis

In elastic wave mechanics, the Fourier transform diagonalizes the wave operator: $\hat{u}(\xi)$ satisfies a simple algebraic equation at each ξ . The FNO exploits the same structure: cardiac PCG signals are periodic, and their physics is most naturally expressed in frequency space.

2. Mode truncation as band-limited inductive bias

In elasticity, we often solve only for the first M eigenmodes of the stiffness matrix (modal reduction). The FNO's truncation to $M = 20$ Fourier modes is exactly analogous: we keep the low-frequency modes where cardiac information lives and discard the high-frequency modes where noise lives.

3. Structural constraints over soft penalties

In nonlinear elasticity, the Legendre-Hadamard (strong ellipticity) condition $Q_{ij}(n) = C_{iA jB} n_A n_B > 0$ can be: (a) enforced as a soft penalty $\max(0, -\lambda_{\min}(Q))$ in the loss, or (b) built structurally into the network output so it always holds. Option (b) is strictly stronger. The PhysioFeatureExtractor is the PCG analog of (b): the periodicity constraint and S1/S2 hierarchy are structural features, not penalties.

4. Physics-guided initialization

In constitutive modeling, initializing a neural network near a known analytical solution (e.g., neo-Hookean: $W = \mu(I_1 - 3)/2$) accelerates convergence and constrains the search space. The S1/S2 mask initialization is exactly this: we start from a physically meaningful filter and adapt it.

5. Learnable physical parameters with physics-based bounds

Material parameters (E, ν, μ) must satisfy physical bounds (e.g., Poisson's ratio $\nu \in (-1, 0.5)$). The sigmoid parameterization of band boundaries ensures they remain in $[0, f_s/2]$ — the analogous physical bound.

15.3 Comparison with PINNs

PINNs enforce PDE residuals in the loss: $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{PDE}}$. The periodicity and hierarchy penalties were analogous: $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda_1 \mathcal{L}_{\text{per}} + \lambda_2 \mathcal{L}_{\text{hier}}$.

But with a crucial difference: in PINNs, the PDE residual is a function of the *network*

output (e.g., $\partial^2 u / \partial t^2 - c^2 \partial^2 u / \partial x^2$ where u is the network’s prediction). Gradients flow freely. In our old design, the penalties were functions of the *input*, not the output. The structural approach (PhysioFeatureExtractor) restores the correct PINN-like behavior: physical quantities are derived from inputs *and* from learnable parameters, so gradients flow into the learnable parameters.

16. Glossary of All Key Terms

Analytic signal

$s_a(t) = s(t) + i\mathcal{H}[s](t)$; complex signal with purely non-negative frequency content; provides envelope and instantaneous phase.

AdamW

Adam optimizer with decoupled weight decay; better regularization for scientific ML models.

Band bump

$\text{sigmoid}(k(f - f_{lo})) \cdot \text{sigmoid}(k(f_{hi} - f))$; differentiable soft indicator for a frequency band.

BatchNorm

Normalizes activations across the batch dimension; stabilizes 2D CNN training.

Cardiac cycle

One complete heartbeat: systole (contraction, ejection) + diastole (relaxation, filling). Duration $\approx 0.6\text{--}1.0\text{ s}$ at rest.

Convolution theorem

$(k * u)(x) \leftrightarrow \hat{k}(\xi) \hat{u}(\xi)$; convolution in time = multiplication in frequency.

Cosine annealing

LR schedule: $\alpha_t = \frac{\alpha_{\max}}{2}(1 + \cos(\pi t/T))$; large steps early, fine steps late.

Cross-beat consistency

Fraction of PSD at HR fundamental + harmonics; measures cardiac cycle regularity.

Cross-entropy loss

$-\frac{1}{B} \sum_b \log p_{y_b}$; equivalent to MLE under categorical distribution.

DFT

Discrete Fourier Transform; maps L time-domain samples to L frequency bins.

Envelope

$e(t) = |s_a(t)|$; instantaneous amplitude of the signal.

FNO

Fourier Neural Operator; learns operators via complex weights on retained Fourier modes; global operator with mode truncation.

GELU

$x\Phi(x)$; smooth activation function used in modern sequence models.

Global average pooling

$\bar{x} = L^{-1} \sum_n x_n$; collapses time dimension; length-invariant.

Gradient clipping

Rescale gradients so $\|g\|_2 \leq 1$; prevents spectral explosion.

Harmonic weights

Softmax-normalized learnable weights on $f_0, 2f_0, 3f_0, 4f_0$; reveal which HR harmonics are most discriminative.

Hilbert transform

Doubles positive-frequency components; produces imaginary part of analytic signal.

Inductive bias

Structural assumption built into model architecture.

InstanceNorm

Per-sample per-channel normalization; robust to amplitude variability.

Instantaneous frequency

$\frac{1}{2\pi} \frac{d}{dt} \arg(s_a(t))$; local oscillation rate; elevated during murmurs.

Logit

$\text{logit}(p) = \ln(p/(1-p))$; unconstrained parameter mapped to $(0, 1)$ via sigmoid.

Mel scale

Perceptual frequency scale: $f_{\text{mel}} = 2595 \log_{10}(1 + f_{\text{Hz}}/700)$.

Mode truncation

Retaining only M lowest Fourier modes; band-limited inductive bias.

Murmur bandwidth

Spectral spread in murmur band; higher in turbulent AS flow.

Nyquist theorem

Faithful sampling requires $f_s \geq 2f_{\text{max}}$.

Operator

Maps a function to a function; generalizes neural networks.

PCG

Phonocardiogram; acoustic recording of heart sounds at chest wall.

Periodicity ratio

Fraction of PSD in cardiac HR band (1–5.3 Hz).

PhysioFeatureExtractor

Computes 18 explicit physical features with learnable band boundaries and harmonic weights.

PhysioFrequencyMask

Learnable soft band-pass: per-mode logits + learnable band boundaries via double-sigmoid bumps.

Power spectral density (PSD)

$P(\xi) = |\hat{s}(\xi)|^2$; energy at each frequency.

rfft

Real FFT; exploits Hermitian symmetry for real signals; output length $L/2 + 1$.

RMS amplitude

$\sqrt{\frac{1}{L} \sum s[n]^2}$; root-mean-square energy.

S1 First heart sound; mitral/tricuspid closure; 25–45 Hz; 100–170 ms.

S2 Second heart sound; aortic/pulmonic closure; 50–70 Hz; 60–100 ms.

S1/S2 energy ratio

E_{S1}/E_{S2} ; encodes relative valve closure intensity.

Spectral centroid

$\sum_k P[k]f_k / \sum_k P[k]$; frequency center of mass.

Spectral leakage

Artificial high-frequency content from abrupt signal truncation; mitigated by Hann windowing.

STFT

Short-Time Fourier Transform; time-frequency representation via overlapping windows.

t-SNE

Non-linear 2D embedding preserving local neighborhood structure; uses Student t -distribution to prevent crowding.

Xavier initialization

Scale weights as $1/\sqrt{C_{\text{in}} \cdot C_{\text{out}}}$; keeps activation variance $O(1)$ at initialization.

17. Further Reading

- Z. Li, N. Kovachki et al., “Fourier Neural Operator for Parametric Partial Differential Equations,” *ICLR* 2021.
- L. van der Maaten & G. Hinton, “Visualizing Data using t-SNE,” *JMLR* 9 (2008), pp. 2579–2605.
- R. Clifford et al., “Classification of Normal/Abnormal Heart Sound Recordings: The PhysioNet/CinC Challenge 2016,” *Computing in Cardiology* 2016.
- I. Loshchilov & F. Hutter, “Decoupled Weight Decay Regularization,” *ICLR* 2019.
- S.M. Raza Ali, “Classification of Heart Sound Signal Using Multiple Features,” *Applied Sciences* 8(12), 2344, 2018.
- M. Raissi, P. Perdikaris, G.E. Karniadakis, “Physics-Informed Neural Networks,” *Journal of Computational Physics* 378, 2019.
- B. Lusch, J.N. Kutz, S.L. Brunton, “Deep learning for universal linear embeddings of nonlinear dynamics,” *Nature Communications* 9, 2018.

- D. Steigmann & M. Shirani, “Cosserat Elasticity of Lattice Shells with Kinematically Constrained Microrotations,” *Mathematics and Mechanics of Solids* 2019.
- B. McFee et al., “librosa: Audio and Music Signal Analysis in Python,” *SciPy* 2015.